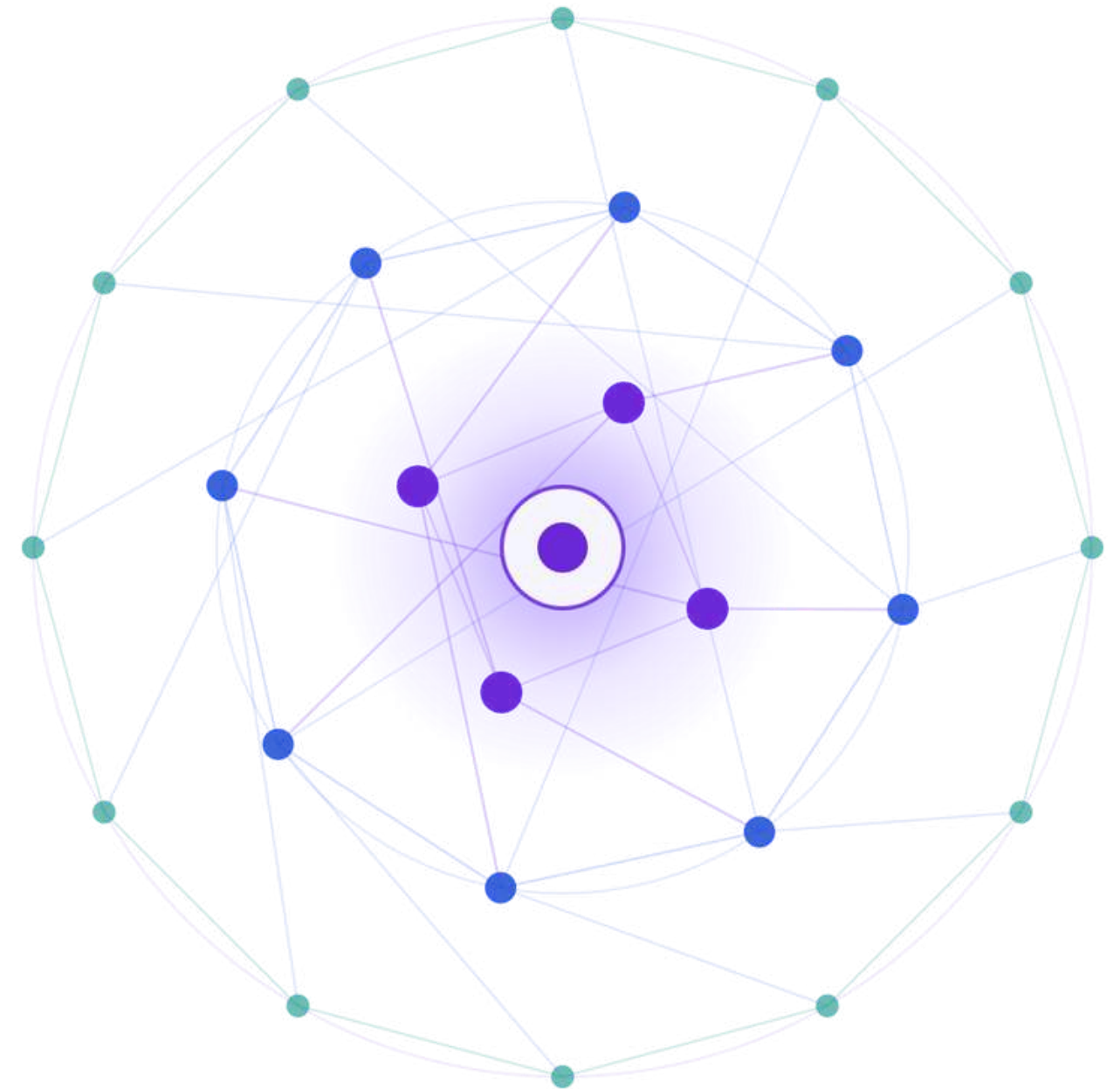


SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

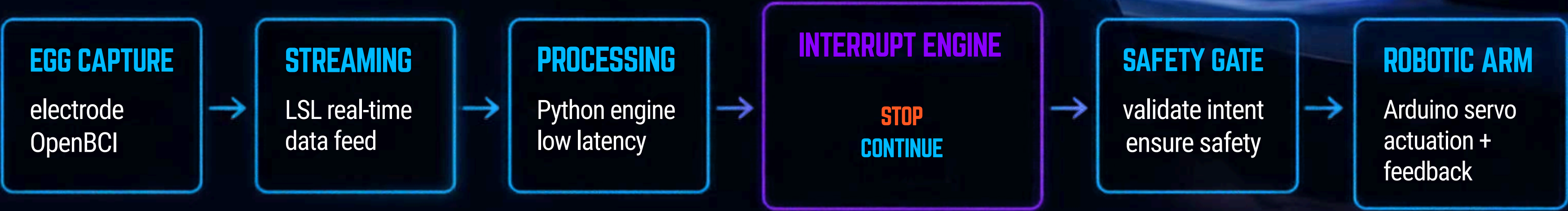
- Theme ID - 05
- Team Name - WalkingDeadlines
- College Name - SRM Institute of Science and Technology
- Member Name & Email 1- Rahul Agarwal (ra2632@srmist.edu.in)
- Member Name & Email 2- Abhinav Sharma (as1040@srmist.edu.in)
- Member Name & Email 3- Anuja Murugan (am6140@srmist.edu.in)
- Member Name & Email 4-
- Submission Github link -



INTERRUPTIBLE REAL-TIME AGENTS

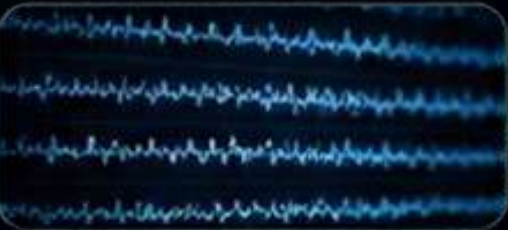
NEUROCONTROL BOT

Detect changing EEG intent during action.
Safely stop, continue, or redirect—without losing context.



EXISTING SOLUTIONS → CRITICAL GAPS → OUR OPPORTUNITY

EXISTING SOLUTIONS TODAY



EEG visualization and recording



Offline signal processing



Joystick or switch based control



Preprogrammed robotic arms



Research BCI prototypes

CRITICAL GAPS



Signal variability across users

Lack of continuous real-time processing

Slow translation from intent to action

Limited hardware integration

Gap between detection and physical action

OUR OPPORTUNITY: NEUROCONTROL BOT

OpenBCI EEG



LSL stream



Python threshold event detection

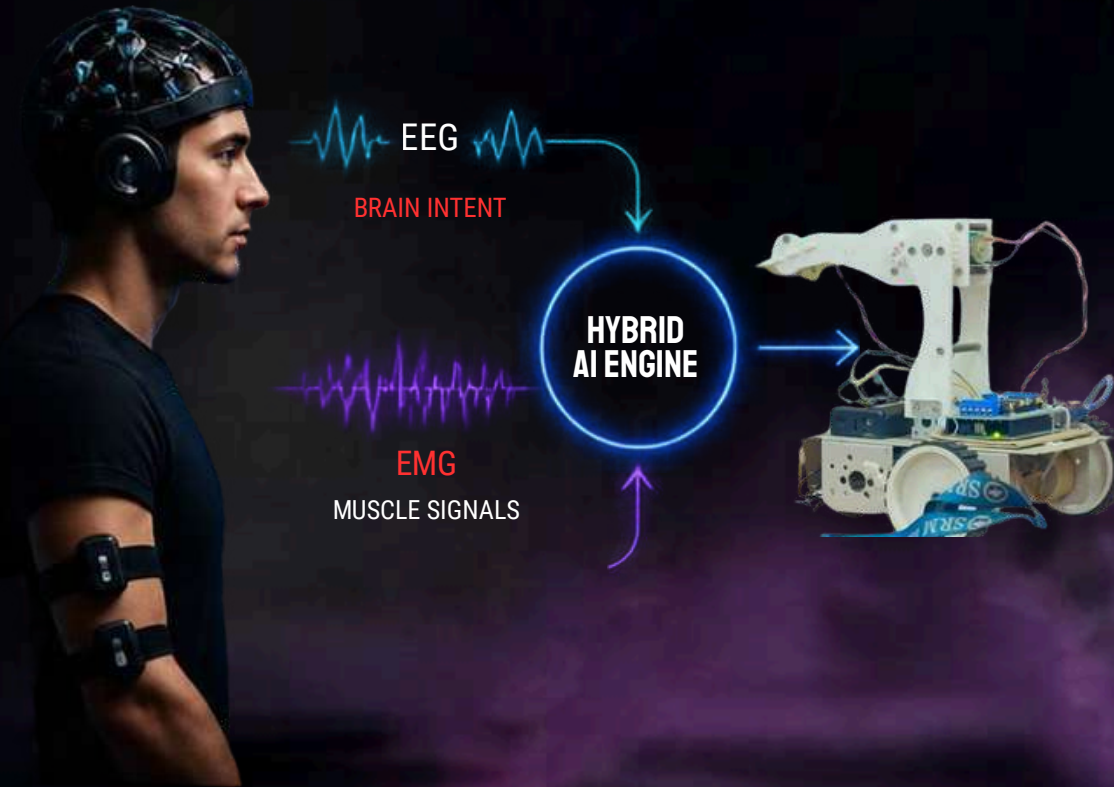


Arduino control



Robotic arm physical action

A HYBRID BRAIN-MUSCLE INTERFACE THAT UNDERSTANDS INTENT AND ACTS



INTUITIVE. ADAPTIVE. AFFORDABLE. CONTROL THAT FEELS NATURAL.

OUR NOVELTY

COMPLETE LOOP
EEG to real physical action

REAL-TIME
Continuous intent to action

LIGHTWEIGHT
Efficient tools and processing

MODULAR
Flexible and scalable system

PHYSICAL PROOF
Working robotic arm demonstration

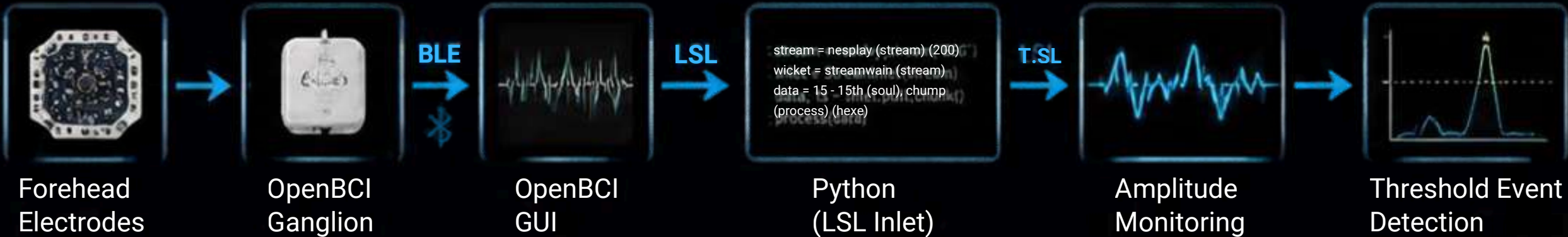
NEUROCONTROL BOT

OUR SOLUTION & ARCHITECTURE

REAL-TIME response
LIGHTWEIGHT & efficient
MODULAR & extensive
INTERPRETABLE & transparent
PHYSICAL proof of concept

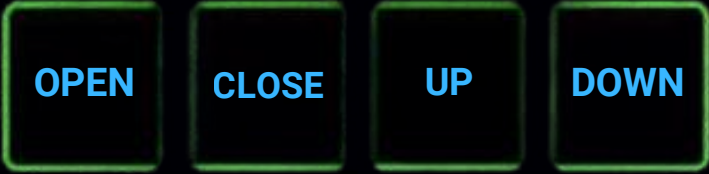
Brain signals to robotic action in real time

SIGNAL ACQUISITION & SOFTWARE PIPELINE

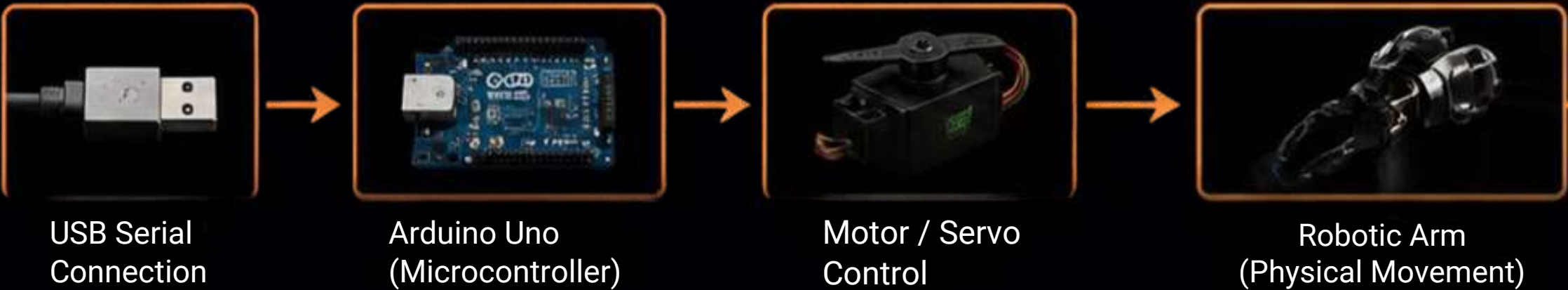


THRESHOLD
CROSSED
EVENT DETECTED

COMMANDMAPPED



HARDWARE & ACTUATION PIPELINE



FUTURE SCOPE

Advanced filtering
Adaptive calibration
More movements
EGG-EMG exploration

NEUROCONTROL BOT

DEMO & PRODUCT WALKTHROUGH

YOUTUBE DEMO VIDEO - [HTTPS://YOUTU.BE/TXFSGJYZOKO](https://youtu.be/txfsgjyzoko)

FROM BRAIN SIGNAL TO ROBOTIC ACTION

FOREHEAD SIGNAL ACQUISITION WITH OPENBCI GANGLION

BLE → OPENBCI GUI → LSL STREAMING

PYTHON AMPLITUDE-THRESHOLD EVENT DETECTION

USB SERIAL COMMAND TO ARDUINO UNO

ROBOTIC ARM EXECUTES OPEN, CLOSE, UP, DOWN

01

USER SETUP
ELECTRODES

02

OPENBCI
GANGLION
LIVE ACQUISITION

03

BLE → GUI → SSL
LIVE STREAMING

04

PYTHON
EVENT DETECTION

05

USB SERIAL
TO ARDUNO

06

ROBOTIC ARM
OPEN, CLOSE,
UP, DOWN



Tools & Tech Stack Used

Mind-Controlled Robotic Arm



BCI & SIGNAL ACQUISITION

electrode EGG / EMOTIV Insight
OpenBCI GUI
MyoWare 2.0. EGM
LSL TimeSeriesRaw EGG



REAL-TIME SOFTWARE

LSL
VS Code
Python
pyls, NumPy, pyserial
Project scripts & agent STOP / CHANGE

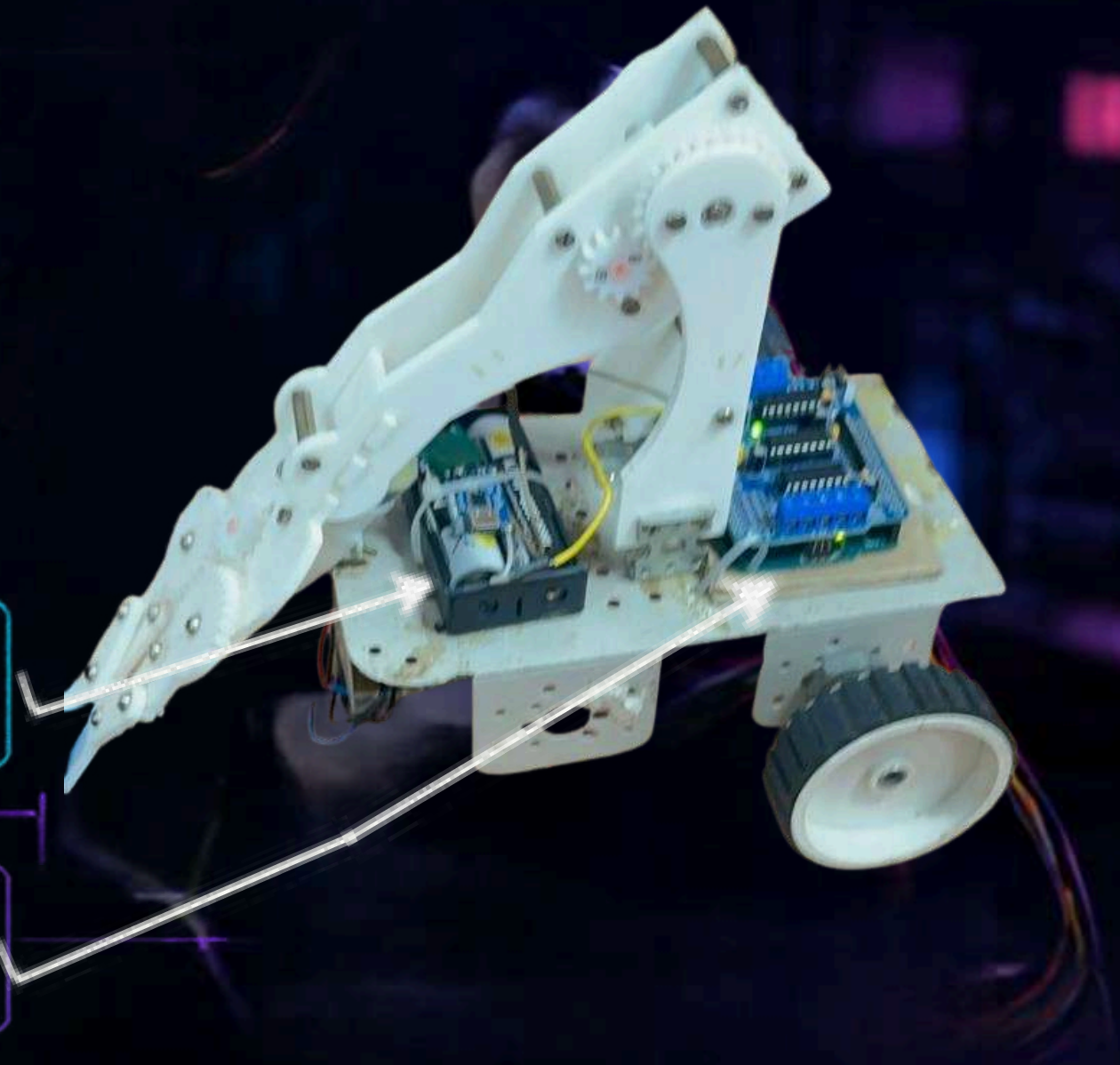


ARDUINO & MOTOR CONTROL

Arduino
Arduino IDE
Serial 9500
AFMotor.h
Adafruit Motor Shield V1
two_mot function.ino
O / C / U / D mapping

Motor 1
Claw Control

Motor 2
Arm Control



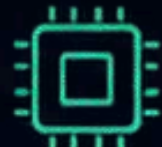
EGG
(EMOTIV Insight)



OpenBCI GUI
LSL TimeSeriesRaw



LSL + Python
Processing



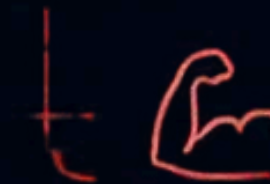
Arduino
(Serial 9600)



Adafruit Motor Shield
V1



Robotic Arm
Actuation



EMG (MyoWare 2.0)
Hybrid Extension

Impact & Use Case

Enabling Independent Living through Brain-Controlled Interaction



IMPACT

01



Greater Independence

Helps individuals with limited mobility perform daily tasks without physical assistance.

02



Improved Quality of Life

Enables natural and intuitive interaction with the environment.

03



Accessible & Affordable

Uses low-cost, non-invasive EGG technology for wider adoption.

04



Real-World Applicability

Can be integrated into assistive devices, rehabilitation tools and smart homes.

05



Inclusive Society

Creates opportunities for physically disabled individuals to live, learn and work independently.



USE CASES

01

Daily Living Assistance

- Pick and place objects
- Open/close containers
- Switch appliances



02

Healthcare & Rehabilitation

- Assist stroke or spinal injury patients
- Support rehabilitation exercises
- Improve motor skill training



03

Assistive Devices

- Wheelchair-mounted arm
- Bedside assistance for mobile patients
- Independent feeding and grooming



04

Industrial & Hazardous Environments

- Remote operation in unsafe areas
- Handling objects in difficult conditions



05

Smart Home Automation

- Control household devices
- Open doors, operate switches
- Hands-free interaction with surroundings



Brain to Action

Intuitive • Real-Time • Non-Invasive



Empowering People

Assist • Support • Enable



Towards a More Inclusive Future

Technology for a Better, Independent Life

INNOVATION HIGHLIGHTS, RESULTS AND LIMITATIONS

NEUROCONTROL BOT

SIMPLE BRAIN SIGNALS → REAL-TIME CONTROL → REAL-WORLD IMPACT



INNOVATION HIGHLIGHTS



Non-Invasive BCI Control

Uses OpenBCI Ganglion with forenzed electrodes for comfortable and easy brain signal acquisition.



Real-Time Brain Intent Detection

Detects natural brain signals and maps them to robotic arm movements with low latency using LSL.



Simple & Affordable Setup

Built with open-source tools (OpenBCI, Python, Arduino), making it cost-effective and scalable.



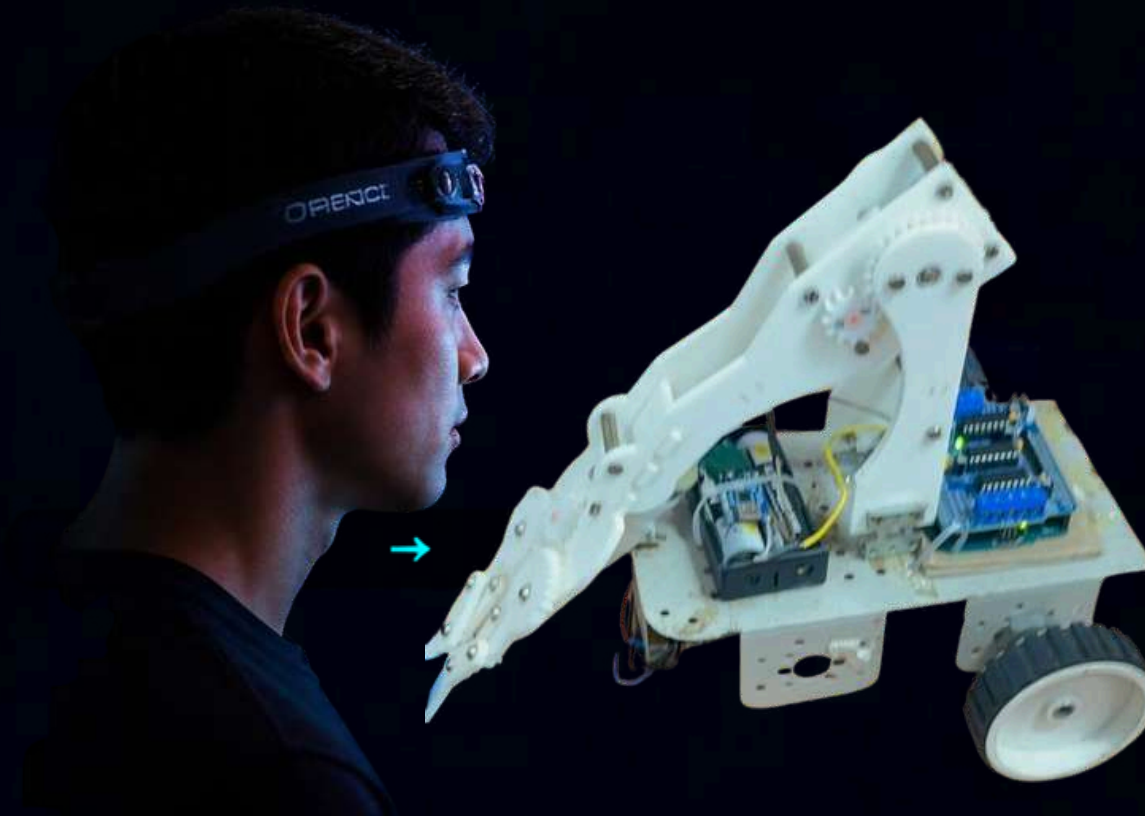
Intuitive & Easy to Use

Controls arm using simple mental intent — Open / Close / Up / Down without extensive training.



Scalable for Future Applications

Can be extended to more movements and integrated into assistive devices, smart homes and industrial applications.



RESULTS



Real-Time Response

200-400 mins
(low latency)



Command Accuracy

~ 90%+
(for defined intents)



Stable Operation

Smooth and
continuous control



Demonstrated Movements

Open | Close
Up | Down



LIMITATIONS



Signal Variability

E6G signals can vary between users and be affected by noise and external factors.



Limited Commands

Currently supports basic movements (Open, Close, Up, Down).



User Calibration

Requires initial setup and threshold adjustment for each user.



Environmental Sensitivity

Performance may be affected by distractions, fatigue and electrode placement.



Not Fully Autonomous

Depends on user intent; advanced functions require further development and training.

NEUROCONTROL BOT

FROM TODAY'S PROTOTYPE
TO TOMORROW'S POTENTIAL

WHAT'S NEXT?

THE NEXT STEPS TO A
SMARTER, MORE CAPABLE
HUMAN-MACHINE INTERFACE

BETTER SIGNAL FILTERING

cleaner, more
stable EEG input

ADAPTIVE CALIBRATION

personalized
thresholds for
each user

MORE MOVEMENTS

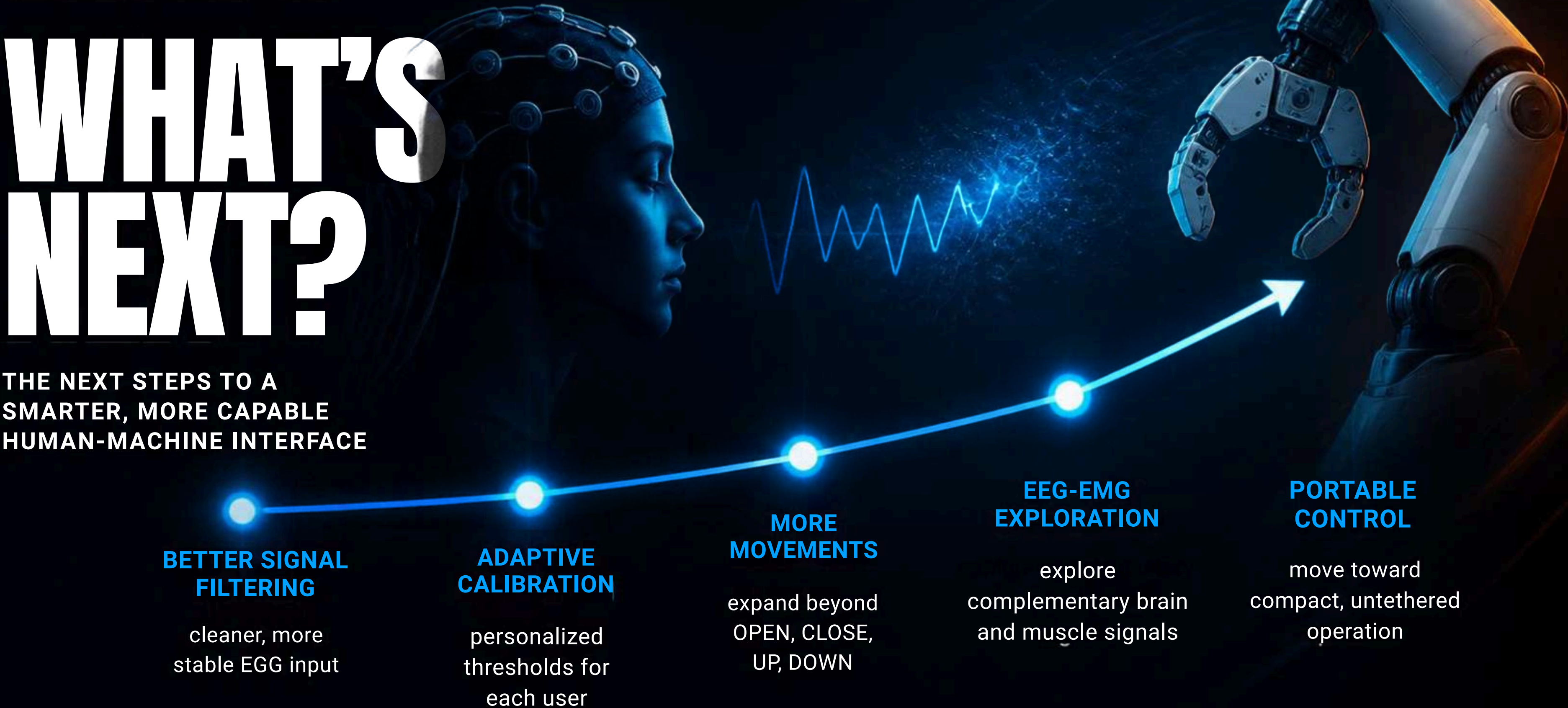
expand beyond
OPEN, CLOSE,
UP, DOWN

EEG-EMG EXPLORATION

explore
complementary brain
and muscle signals

PORTABLE CONTROL

move toward
compact, untethered
operation



BROWNIE POINTS

WHY NEUROCONTROL BOT STANDS OUT

NON-INVASIVE EEG

10-electrode
OpenRii setup
captures brain
signals without
surgery.

REAL-TIME PIPELINE

LSI streams EEG
continuously to
Python for low-
latency processing.

INTERRUPTIBLE CONTROL

Intent changes
can trigger
STOP, CHANGE
or CONTINUE
during execution.

SAFETY-FIRST ACTION

A threshold and
safety gate
validates
commands before
robotic movement.

AFFORDABLE & MODULAR

Open-source
software
and Arduino-based
control keep the
prototype extensive.

IMPACT

Greater independence
Natural hands-free interaction
Safer interruptible execution

USE CASES

Reach, grasp, pick-place
Rehabilitation research
Limited-mobility assistive control
Human-Robot Interaction experimentation



Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo (Y/N)
- README with reproducible setup instructions (Y/N)
- Demo video, max 5 minutes (**You**Tube or Drive link)
- Presentation file (PPT or PDF) (Y/N)

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India